Development and validation of an automated algorithm for identifying patients at high risk for drug-induced hypoglycemia

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Purpose. Hypoglycemia is one of the most concerning adverse drug events in hospitalized patients. Using information from institutional electronic health records, we aimed to develop dynamic predictive models to identify patients at high risk for hypoglycemia during antihyperglycemic therapy.

Methods. The study population consisted of 21,840 patients who received antihyperglycemic medication on any of the first 5 hospital days (the “risk model days”) at 2 large hospitals. Data on candidate predictors were extracted from discrete electronic health record fields to construct models for predicting hypoglycemia within 24 hours after each risk model day. Final models were internally validated by replication in 100 bootstrap samples and reapplying model parameters to the original risk population.

Results. The development and validation sample included 60,762 risk model days followed by 1,256 days with hypoglycemic events (2.07 events per 100 risk model days). The days 3, 4, and 5 models presented similar associations between predictors and the risk of hypoglycemia and were therefore collapsed into a single model. The strongest hypoglycemia risk factors across all 3 risk periods (day 1, day 2, and days 3–5) were blood glucose (BG) fluctuations, BG trend, history of hypoglycemia, lower body weight, lower creatinine clearance, use of long-acting or high-dose insulin, and sulfonylurea use. C statistics for the 3 models ranged from 0.844 to 0.887. Depending on the model used, risk scores in the upper 90th percentile predicted 48.5–63.1% of actual hypoglycemic events. It was estimated that by targeting only patients in the upper 90th percentile, providers would need to intervene during fewer than 9 admissions to prevent 1 hypoglycemic event.

Conclusion. The developed prediction models were found to have excellent discriminative validity and good calibration, allowing clinicians to focus interventions on a select high-risk population in which the majority of hypoglycemic events occur.

Keywords: antidiabetic drugs, electronic health records, hypoglycemia, prediction model, risk score

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Over one fourth of hospitalized patients in the United States have diabetes, positioning blood glucose (BG) management as a prevalent and challenging task in inpatient settings.[1,2] While aggressive BG control over even short time periods reduces the risk of complications, including death, concerns about hypoglycemia remain a limiting factor.[3] Hypoglycemia is associated with increased costs, hospital length of stay, and mortality and morbidity attributable to cardiovascular, cerebrovascular, and fall events.[4,5] The National Action Plan for Adverse Drug Event Prevention reports that about one third of medication error–related deaths and one quarter of instances of patient harm are due to challenges in insulin administration.[6]

While hospital settings allow for much tighter calibration of glucose management, they also pose additional challenges relative to outpatient treatment, including acute changes in clinical presentation that influence glucose levels, related changes in drug regimens and nutritional intake, and the interplay of these factors. Risk factors associated with hospital-acquired hypoglycemia include aggressive dosing or inappropriate timing of insulin therapy, concomitant use of medications with hypoglycemic effects, inadequate nutritional intake, decreased renal function, failure to consider a patient's history of hypoglycemia, insufficient BG monitoring, and unmeasured dynamic changes in the aforementioned factors.[3,7-15]

Because of the need to reevaluate and adjust glucose management strategies consistently during hospitalization, automated risk prediction algorithms may enhance providers' recognition of emerging hypoglycemia risk and aid in prioritizing patients for preventive interventions. A number of algorithms involve the use of continuous glucose monitoring (CGM) systems, which generate real-time alerts when hypoglycemia is predicted.[16-22] Widespread adoption of CGM by hospitals, however, has been limited by insufficient data on its cost-effectiveness and safety in the inpatient setting.[23,24] In the absence of CGM, increased monitoring of sporadically measured BG values and other measurable risk factors may allow prediction of hypoglycemic risk.

Several research teams have developed risk prediction algorithms that are specific to inpatient settings.[4,25,26] These algorithms rely on clinical data available at the time of admission and do not provide for dynamic evaluation of risk factor changes throughout a hospital stay. Accordingly, their predictive validity has been moderate, with reported C statistics of 0.70–0.73. In the study described here, we aimed to construct and validate a dynamic electronic health record (EHR)–based prediction model targeting patients receiving antihyperglycemic therapy in hospitals to assist with early identification of hypoglycemia risk.

KEY POINTS

- Using discrete data from fully integrated electronic health records, dynamic risk-scoring models for predicting inpatient hypoglycemia were developed and shown to have excellent discriminative performance.
- By using the models to target only admissions in the upper 90th percentile for risk score, providers would need to intervene in fewer than 9 admissions to prevent 1 hypoglycemic event.
- In model testing, the strongest risk factors for hypoglycemia were blood glucose (BG) fluctuations, BG trend, history of hypoglycemia, lower body weight, lower creatinine clearance, long-acting or high-dose insulin use, and sulfonylurea use.

Methods

Study design. The study was part of a larger project aimed at developing an EHR-based score for ranking hospitalized patients according to their risks of 16 adverse drug events. The project, study populations, and data sources have been described in detail elsewhere.[27] In brief, we identified a retrospective cohort of hospitalized patients using inpatient EHR data from the 2 largest University of Florida (UF)–affiliated hospitals; data were collected from UF Health Shands Hospital for the period January 2012–October 2013 and from UF Health Jacksonville for the period March 2013–October 2013. We developed 5 individual prediction models corresponding to the first 5 hospital days (the "risk model days"), which were used to predict the risk of hypoglycemia on the following day. We opted for individual prediction models to allow analysis of changes in risk factors and their association with hypoglycemia as patients progress through a hospital stay. Later hospital days were omitted because of small sample sizes.

Study population. We considered all patients 18 years of age or older on admission from all admitting services. To be included in the cohort for a particular risk model day, patients must have received at least 1 oral antihyperglycemic agent (except acarbose or miglitol) or insulin; this criterion was stipulated to focus our prediction modeling on drug-induced hypoglycemia (Appendix A).

Outcome definition. We defined hypoglycemia as a serum BG value of <50 mg/dL that was not followed by a BG value of >80 mg/dL within 10 minutes (this approach was intended to confirm the validity of initially measured low BG values).

Hypoglycemia risk factors. We conducted a comprehensive literature search to identify risk factors for hospital-acquired hypoglycemia, with a focus on patient safety literature and risk factors that reflect quality deficits in glycemic management.[3,7-13] The list of risk factors, or candidate predictors, was augmented by drug monograph reviews, clinical expert opinion, and discussion with a National Technical Expert Panel that was assembled to support the project.

Selection of candidate predictors was limited to those that are measurable from discrete fields in standard EHRs and have reasonable face validity with the intention of developing standardized definitions that can be applied in different EHRs and hospital systems. Each risk factor was measured during a risk factor–specific look-back period in order to ensure sufficient proximity to development of hypoglycemia (Appendix B). Risk factor information was collected up to midnight of a given hospital day, and hypoglycemia manifestation was evaluated during the following 24 hours. We chose this time sequence to mimic prospective implementation of the risk scoring method, whereby providers would receive a prioritized list of high-risk patients at the beginning of the morning shift. Missing data for a risk factor were accounted for either by entry of a missing-data indicator (if missingness had predictive value on its own) or by simple imputation (i.e., replacing the missing value with an extrapolated value such as a mean).

Data analysis

We conducted descriptive analyses for 64 risk factor candidates and assessed their univariate associations with hypoglycemia. Clinical experts validated risk factor definitions via chart review and defined data cleaning steps (e.g., to remove extraneous values) as necessary. To reduce the chance of model overfitting, we conducted cluster analyses to identify highly correlated risk factors and retained a risk factor within a given cluster according to the factor's strongest contribution to model fit (measured per the Akaike information criterion [AIC]). We further evaluated the predictive properties of various transformations of each continuous variable (e.g., log, restricted cubic spline)[28] in univariate analysis and chose the transformation with the best AIC value.

We constructed risk models for days 1 and 2; a third model was constructed by collapsing days 3–5 given that risk factor associations with hypoglycemia remained unchanged across those 3 days. Because clusters of multiple observations created in the collapsed model may be correlated, we compared the parameter estimates derived from simple logistic regression, assuming independence, and those derived using generalized estimating equations accounting for correlation. Since the clustering had only marginal effects on parameter estimates (results not shown), for simplicity we retained logistic regression as the method of analysis.

For each day or day cluster, we examined 4 different analytic models: a nonparsimonious model including all risk factors, an expert model that retained risk factors considered most relevant by clinical experts, a parsimonious model that used fast backward elimination and retained all risk factors with an association that had a p value of <0.15, and a reduced backstep model including all risk factors that were retained at the same p value in at least 40 of 100 backward elimination processes applied to 100 bootstrap samples derived from the original model cohort. We chose the best model on the basis of 3 performance indicators: discrimination (as measured by the C statistic), goodness of fit (as measured by the AIC value), and calibration (as measured by the Hosmer–Lemeshow goodness-of-fit test).[29]

Finally, the selected model for each risk model day or day cluster was validated through replication using 100 bootstrap samples derived from the original cohort. The original model was considered valid if the validated C statistic was located within the 95% confidence interval (CI) of the C statistic derived from the original data set.[29] All statistical analyses were conducted using SAS software, version 9.4 (SAS Institute, Cary, NC), and R software, version 3.1.1 (R Core Team, 2014; R Foundation for Statistical Computing, Vienna, Austria). The study was approved by the UF institutional review and privacy boards.

Results

Our sample included 21,840 admissions across hospital days 1–5 during which patients received antihyperglycemic medications on 60,762 risk model days. Collected data on those admissions were used to predict 1,256 days with a hypoglycemic event (20.7 events per 1,000 risk model days). Hypoglycemia risk dropped slightly during the examined days, with rates of 2.4% (294 events per 12,264 risk model days), 2.3% (358 per 15,810), and 1.8% (604 per 32,688) for risk model days 1, 2, and 3–5, respectively (Figure 1).

About 80% of the risk model population had a diagnosis of diabetes mellitus (10% had type 1 diabetes), and 26% had a history of hypoglycemia during a previous inpatient or outpatient encounter. A majority of patients received insulin, and less than 10% received sulfonylureas (Table 1). Approximately 60% used medications without an indication for diabetes but with hypoglycemic effects.

Candidate variable selection. Among 64 variables that we reviewed, more than 20 variables were excluded after univariate analyses. For example, we attempted to measure decreased oral intake by assessing the nurse-recorded proportion of the food tray eaten. However, that variable was excluded due to unreliable documentation that resulted in limited predictive value. Also, the small number of hypoglycemic events documented in the data set limited our ability to evaluate several insulin dosing–related risk factors (e.g., rapid increase in insulin dose, excessive use of single-dose correction insulin) and specific drug products (e.g., opioids, quinolones, pentamidine).

Predictive model evaluation and validation. The reduced backstep models were found to have the best performance across all days and were chosen as the final models. These models had strong discriminative validity, with C statistics of 0.844 (95% CI, 0.824–0.864), 0.874 (95% CI, 0.858–0.891), and 0.887 (95% CI, 0.874–0.899) for days 1, 2, and 3–5, respectively (Figure 2). All C statistics were successfully validated via the internal validation procedure (C values were 0.831, 0.862, and 0.878 for days 1, 2, and 3–5, respectively). Model calibration was good (Appendix C), as indicated by nonsignificant Hosmer–Lemeshow goodness-of-fit statistics

($p = 0.4872$, $p = 0.0807$, and $p = 0.0913$ for days 1, 2, and 3–5, respectively).

Figure 1 summarizes the predictive characteristics of the models. Hospital admission with a risk score in the upper 50th percentile predicted 278 of 6,133 (4.5%), 340 of 7,906 (4.3%), and 580 of 16,345 (3.5%) of all hypoglycemic events documented during 24-hour periods after days 1, 2, and 3–5, respectively. Among admissions in the 90th percentile of the risk score distribution, 143 of 294 (48.6%), 211 of 358 (58.9%), and 381 of 604 (63.1%) hypoglycemic events were captured by the prediction models, with better model performance on later hospital days (see Appendix C for the percentage of total hypoglycemia events in each risk score decile). If only admissions in the upper 90th percentile of the risk score distribution were targeted, providers would need to intervene during approximately 9 admissions to prevent 1 hypoglycemic event.

Hypoglycemia risk factors. Table 2 characterizes the multivariable associations between hypoglycemia and individual predictors across the 3 models (applicable to days 1, 2, and 3–5). For demographics, male sex was a significant predictor of hypoglycemia only in the day 1 model. Age at admission was not associated with hypoglycemia after adjustment for all other risk factors. In all models, there was an inverse linear association between body weight and the risk of hypoglycemia, with a 1–3% reduction in the odds of hypoglycemia for every 1-kg increase in weight. Fever on day 1 increased the risk of hypoglycemia on day 1 only (odds ratio [OR], 1.22; 95% CI, 1.00–1.48). Patients who received antihyperglycemic treatment in an intensive care unit (ICU) during hospital days 3–5 were less likely to develop hypoglycemia.

Among all evaluated metrics of glucose control, BG fluctuations, BG trend, and history of hypoglycemia were the most consistent predictors of hypoglycemia across all model days, with ORs ranging from 1.55

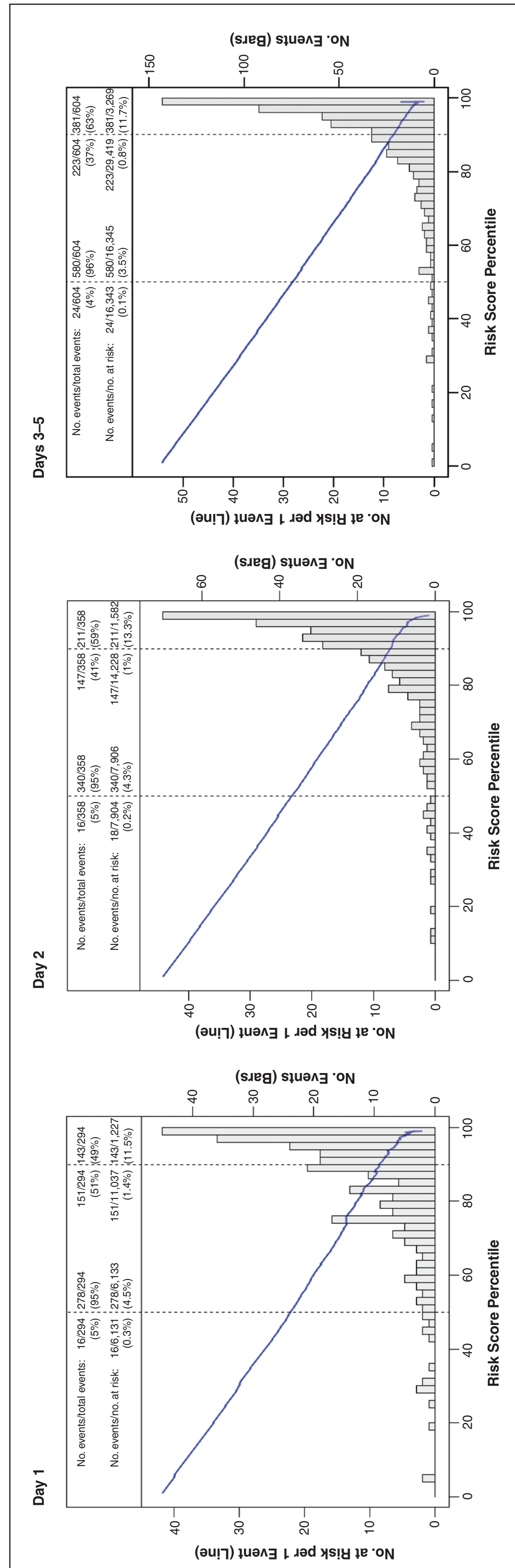

Figure 1. Hypoglycemia rates and number at risk per 1 event by risk score percentile (at top of each panel) for the 3 modeled risk periods. The gray bars indicate the number of hospital days with hypoglycemia events per risk score decile. The blue line represents the number of admissions flagged during a risk model day that would require intervention in order to prevent 1 hypoglycemia event on the subsequent day.

Table 1. Clinical Characteristics of Risk Model Patient-Days With and Without Hypoglycemia on Subsequent Day[a]

Variable	No. (%) Risk Model Days Without Hypoglycemia[b] (*n* = 59,506)	No. (%) Risk Model Days With Hypoglycemia[b] (*n* = 1,256)
Demographics Related		
Male sex	29,486 (49.6)	552 (44.0)
Mean ± S.D. age on admission, yr	59.82 ± 15.19	55.22 ± 17.4
BG Related		
History of hypoglycemia within 1 yr of admission	15,339 (25.8)	741 (59.0)
Mean ± S.D. maximum BG value	242.8 ± 98.39	276.0 ± 136.5
Mean ± S.D. minimum BG value	131.3 ± 54.46	97.5 ± 65.0
Mean ± S.D. admission BG	189.7 ± 112.2	228.6 ± 149.6
Mean ± S.D. BG trend[c]	0.1 ± 5.4	−1.0 ± 8.9
Mean ± S.D. BG fluctuation[d]	111.4 ± 84.5	178.5 ± 123.7
Use of dextrose 50 mg	2,051 (3.5)	211 (16.8)
Antihyperglycemic Drug Related		
Initiation of new antidiabetic agent	11,353 (19.08)	160 (12.74)
Use of insulin	56,191 (94.4)	1,203 (95.8)
Long-acting	22,633 (38.0)	912 (72.6)
Subcutaneous	54,495 (91.6)	1,195 (95.1)
Infusion	7,096 (11.9)	169 (13.5)
Use of sulfonylureas	4,157 (7.0)	108 (8.6)
Use of nonantidiabetic agents with hypoglycemic effects[e]	30,411 (51.1)	755 (60.1)
Oral Intake Related		
Nausea	11,323 (19.0)	255 (20.3)
Active order for nothing by mouth	9,412 (15.8)	207 (16.5)
Enteral feeding	8,579 (14.4)	76 (6.1)
Service Location Related		
Acute care unit	45,870 (77.1)	1,120 (89.2)
Intensive care unit	9,835 (16.5)	90 (7.17)
Operating room	3,801 (6.39)	46 (3.7)
Laboratory Value Related		
Low serum albumin (<3.5 mg/dL)	16,309 (27.41)	443 (35.27)
Low body weight (<50 kg)	5,600 (9.41)	234 (18.63)
Most recent CL_{cr} value < 45 mL/min	16,327 (27.4)	558 (44.4)
Hypothermia (body temperature < 36 °C)	11,810 (19.9)	312 (24.8)
Comorbidity Related		
Diabetes mellitus	47,822 (80.4)	1,169 (93.07)
Type 1 diabetes mellitus	5,708 (9.6)	402 (32.0)
Ischemic heart disease	14,862 (25.0)	315 (25.1)

[a]BG = blood glucose, CL_{cr} = creatinine clearance.
[b]Unless otherwise indicated.
[c]BG trend was defined as the slope of the least squared regression line of all recent BG values up to 4 days prior to a risk model day. The regression was fit to a minimum of 4 and a maximum of 10 values.
[d]BG fluctuation was defined as the difference between the maximum and minimum BG values at a risk model day.
[e]Nonantidiabetic drugs with hypoglycemic effects included aripiprazol, azithromycin, carvedilol, codeine, doxepin, gabapentin, haloperidol, hydromorphon, ibuprofen, iron, lanreotide, lorcaserin, mecasermin, metreleptin, micafungin, pancrelipase, pasireotide, pentamidine, piperacillin, pregabalin, quetiapine, sunitinib, tromethamine, and venlafaxine

to 4.52. Of note, to characterize the nonlinear associations between the risk of hypoglycemia and BG trend, maximum BG value, and admission BG value, we used restricted cubic splines (Appendix D), which do not allow straightforward interpretation of ORs. For the day 1 model, BG fluctuation and the trend in BG control drove the prediction of hypoglycemia risk. As expected, a negative BG trend (i.e., decreasing BG values) on day 1 increased the risk of hypoglycemia, which was also true in later-day models. In the day 2 model, in addition to the dynamic changes in BG values, both the admission BG and maximum BG values contributed to hypoglycemia prediction. Low admission BG levels were protective against hypoglycemia. Finally, both maximum and minimum BG values contributed to hypoglycemia prediction in the day 2 and days 3–5 models, with both values demonstrated to have an inverse relationship with hypoglycemia risk (i.e., lower values during the prediction day or period increased the risk of hypoglycemia on the following day). A history of hypoglycemia, either in the year before admission or during admission, was a significant risk factor in all risk models.

Use of long-acting insulin, use of sulfonylureas, use of high-dose subcutaneous insulin, and use of insulin infusions were significant risk factors for hypoglycemia on all model days. When the risk prediction models were adjusted for use of these high-risk regimens, insulin use itself was not found to confer additional risk. The use of nondiabetic medication with hypoglycemic effects was weakly associated with hypoglycemia risk on day 2 (OR, 1.14; 95% CI, 1.01–1.29) and was dropped from the models for day 1 and days 3–5. A recent change in the antihyperglycemic regimen, which indicated (on day 1 exclusively) a switch between continuous insulin infusion and subcutaneous insulin therapy, was a protective factor in the day 1 model (see definitions of variables in Appendix A). Finally, transi-

Figure 2. Area under the curve (AUC) values (with 95% confidence intervals) derived from area under the receiver operating characteristic curve analyses for the 3 modeled risk periods. In each panel, the dashed line represents the line of no discrimination.

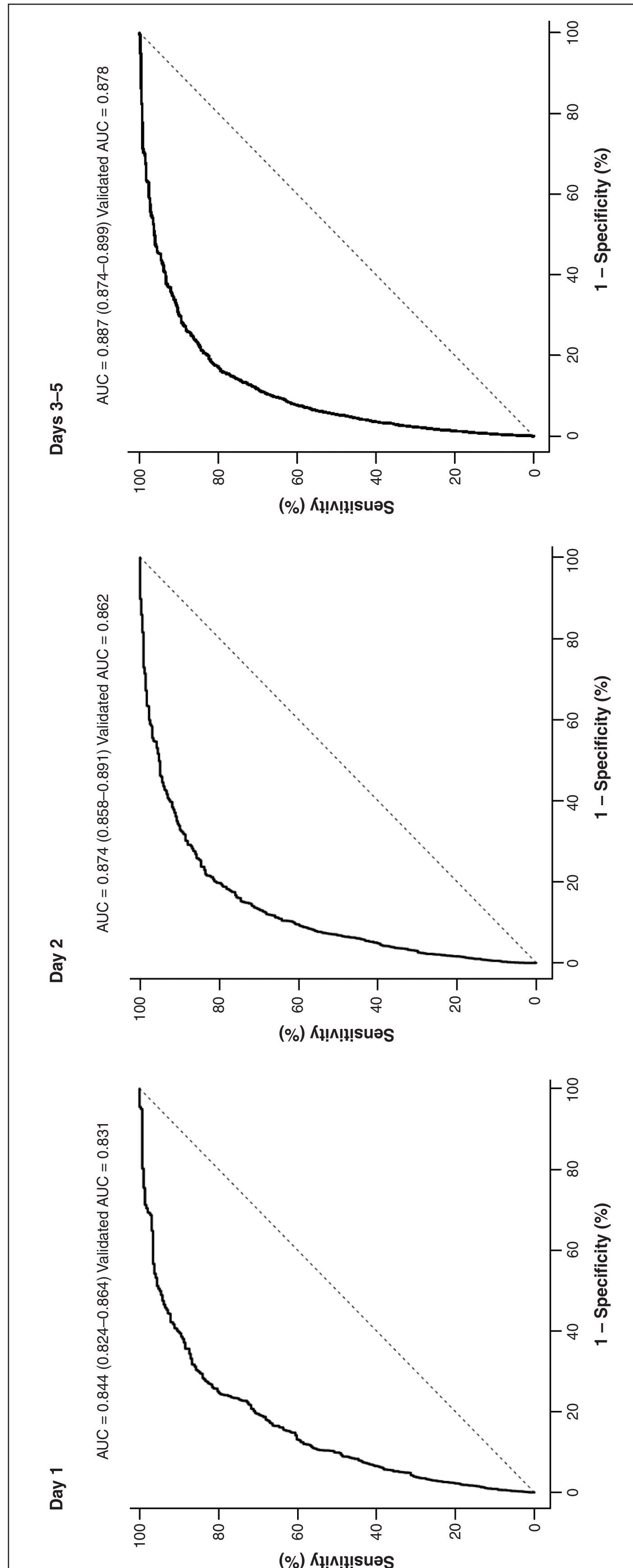

Table 2. Association of Candidate Predictors With Hypoglycemia Events, by Risk Period[a]

Candidate Predictor	Adjusted Odds Ratio (95% CI)[b]		
	Day 1 294 events in 12,264 risk model days (event rate, 2.4%)	Day 2 358 events in 15,810 risk model days (event rate, 2.3%)	Days 3–5 604 events in 32,688 risk model days (event rate, 1.8%)
Basic Characteristics			
Male sex	1.40 (1.09–1.79)	0.82 (0.65–1.04)	Dropped
Age on admission	Dropped	Dropped	Dropped
Body weight	0.97 (0.97–0.98)	0.99 (0.98–0.99)	0.99 (0.98–0.99)
Body temperature	1.22 (1.00–1.48)	0.88 (0.73–1.06)	Dropped
ICU stay	Dropped	Dropped	0.69 (0.49–0.98)
BG Values and History of Hypoglycemia			
Minimum BG value	Dropped	Dropped	0.99 (0.99–1.00)
Maximum BG value	Dropped	Spline	Dropped
Admission BG value	Dropped	Spline	Spline
BG fluctuation (log transformed)	1.59 (1.22–2.07)	4.52 (3.16–6.47)	1.55 (1.20–1.99)
BG trend	Spline	Spline	Spline
No. prior days with BG of <70 mg/dL	. . .[c]	1.27 (0.98–1.64)	1.79 (1.62–1.98)
History of hypoglycemia	2.75 (2.05–3.69)	1.72 (1.33–2.23)	1.68 (1.36–2.07)
In-Hospital Use of Antihyperglycemic Drugs			
Use of insulin (any type)	Dropped	0.55 (0.29–1.06)	0.32 (0.19–0.56)
Insulin infusion	2.85 (1.76–4.62)	1.34 (0.90–1.99)	1.64 (1.14–2.34)
High-dose subcutaneous insulin	1.42 (0.85–2.36)	1.50 (1.05–2.15)	1.32 (1.02–1.70)
Long-acting insulin	3.46 (2.56–4.66)	3.64 (2.61–5.08)	4.64 (3.57–6.04)
Use of sulfonylureas	1.92 (1.16–3.33)	2.10 (1.37–3.23)	2.04 (1.44–2.89)
New initiation of antihyperglycemic drug	Dropped	Dropped	1.39 (0.96–2.01)
No. nonantidiabetic agents with hypoglycemic effects	Dropped	1.14 (1.01–1.29)	Dropped
Change in diabetes regimen	0.31 (0.16–0.60)	1.30 (0.96–1.77)	Dropped
BG Monitoring			
Insufficient BG monitoring for long-acting insulin regimen	Dropped	1.82 (0.91–3.64)	Dropped
No. recorded BG values	Dropped	1.15 (1.05–1.26)	1.31 (0.96–1.78)
Change in Oral Intake			
Nausea	Dropped	0.67 (0.49–0.90)	Dropped
Nothing-by-mouth order	Dropped	Dropped	1.38 (1.07–1.79)
Insulin and presence of an active order for nothing by mouth	Dropped	2.14 (1.36–3.38)	Dropped
Long-acting insulin and presence of an active order for nothing by mouth	Dropped	0.55 (0.32–0.97)	Dropped
Use of dextrose 50%	2.55 (1.78–3.65)	1.42 (0.98–2.04)	Dropped
Enteral feeding	Dropped	Dropped	0.75 (0.52–1.07)

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Table 2. Association of Candidate Predictors With Hypoglycemia Events, by Risk Period[a]

Candidate Predictor	Adjusted Odds Ratio (95% CI)[b] Day 1 294 events in 12,264 risk model days (event rate, 2.4%)	Day 2 358 events in 15,810 risk model days (event rate, 2.3%)	Days 3–5 604 events in 32,688 risk model days (event rate, 1.8%)
Laboratory Values			
Serum albumin	0.60 (0.48–0.76)	Dropped	0.84 (0.72–0.98)
Serum albumin value missing	0.16 (0.07–0.36)	Dropped	0.47 (0.27–0.81)
CL_{cr}	1.00 (0.99–1.00)	0.99 (0.98–0.99)	0.99 (0.99–1.00)
CL_{cr} value of >100 mL/min	0.43 (0.28–0.67)	0.41 (0.28–0.60)	0.50 (0.37–0.67)
CL_{cr} value missing	0.90 (0.51–1.60)	0.50 (0.28–0.91)	0.51 (0.30–0.87)
Other Risk Factors			
Ischemic heart disease	0.94 (0.71–1.24)	Dropped	0.77 (0.62–0.95)
Type 1 diabetes mellitus	1.31 (0.96–1.79)	Dropped	1.43 (1.15–1.78)
Transition between ACU and ICU	0.31 (0.15–0.62)	0.41 (0.19–0.86)	Dropped
On day 3	. . .	. . .	Reference
On day 4	. . .	. . .	0.73 (0.59–0.90)
On day 5	. . .	. . .	0.73 (0.59–0.92)

[a]CI = confidence interval, ICU = intensive care unit, BG = blood glucose, ACU = acute care unit, CL_{cr} = creatinine clearance. "Dropped" denotes variables dropped from final model during backward elimination due to lack of significant association with hypoglycemia. "Spline" denotes variables modeled as restricted cubic spline.

[b]For continuous variables, odds ratios indicate odds of 1-unit change (e.g., each 1-kg increase in body weight reduced the odds of hypoglycemia on risk model day 1 by 3%).

[c]Not applicable; factor not considered for inclusion in model for a given risk period.

tions between hospital units were protective, likely because of potential interruptions in antihyperglycemic therapy.

We found a weak association between hypoglycemia and the use of long-acting insulin without sufficient nighttime monitoring, as well as a significant association between the number of BG values recorded daily and hypoglycemia risk. Surprisingly, use of antinausea medications and use of enteral nutrition were protective against or not significantly predictive of hypoglycemia, as was the use of long-acting insulin during an active order for a patient to receive nothing by mouth when the prediction model was adjusted for use of any insulin during an active order for nothing by mouth. Use of dextrose 50% i.v. was a significant risk factor, likely because it was indicative of previous hypoglycemia.

Finally, a low serum albumin value was a significant risk factor for hypoglycemia, while a missing albumin value was protective. Likewise, lower creatinine clearance (CL_{cr}) was a significant risk factor for hypoglycemia, while a CL_{cr} value greater than 100 mL/min and a missing CL_{cr} value were both protective.

Discussion

To our knowledge, the models described here are the first EHR-based prediction models for identifying patients at high risk for in-hospital drug-induced hypoglycemia. This is also the first study to operationalize and fully automate data collection for all risk factors from discrete EHR fields for near real-time prediction. The modeling method was found to have strong discriminative validity, allowing the capture of more than half of all hypoglycemia events during the 10% of admissions associated with the highest hypoglycemia risk scores. Calibration of the risk models was equally good, as evidenced by accurate prediction of hypoglycemia incidence rates among almost all risk scoring deciles.

We identified 3 previous studies in which researchers developed hypoglycemia risk prediction algorithms for inpatient settings.[4,25,26] The predictive performance of the algorithms was moderate, with reported C statistics of 0.70–0.73, and inferior to our models' performance, owing to several features. First, while the previously described algorithms relied on clinical data available at the time of admission, we constructed dynamic models that updated variable information or introduced additional variables that measured change as patients progressed during their hospital stay. Second, we carefully characterized risk factor associations with hypoglycemia, using transformations including restricted cubic splines. Third, we took advantage of the informative value of missingness, which indicated, for example, that clinicians had foregone the ordering of albumin or serum creatinine tests, likely because of a lack of concern that values could be abnormal. Importantly, risk score development was confined to patients who received antihyperglycemic treatment, which, besides food intake, is the most relevant risk factor for hypoglycemia; in addition, our models focused on severe hypoglycemia, defined as a BG concentration of 50 mg/mL.

Because of the risk of provider alert fatigue surrounding exposure to false-positive clinical decision support messages, priority lists based on risk scores must carefully balance sensitivity and specificity. To illustrate, in our study the upper risk score decile captured more than half of all actual hypoglycemia events in the evaluated sample of admissions. The frequency of hypoglycemia in this decile was above 10%; therefore, the number of patients predicted to be at risk was less than 9 per 1 actual event. In contrast, a focus on admissions in the upper half of the risk score distribution would capture almost all hypoglycemic events but increase the number of patients flagged for intervention by 5-fold (25 patients at risk per 1 event). Importantly, because risk scoring by our method is confined to patients who have received antihyperglycemic treatment, in actual practice the proportion of adult hospital admissions evaluated for intervention would be reduced accordingly (approximately 20% of adult admissions at the 2 study hospitals meet that criterion) and would be further reduced by applying the risk score cutoff used for priority ranking.

This study found a multitude of risk factors associated with hypoglycemia during the first 5 inpatient days, including some deserving further discussion. First, both fluctuation in BG values (i.e., the difference between peak and trough values on a risk model day) and the BG trend (as estimated over up to 4 previous days) were consistently predictive of hypoglycemia, while the strength of associations between single BG values (e.g., minimum or maximum level) and hypoglycemia risk was variable. While the risk of hypoglycemia associated with a downward BG trend is intuitive, extensive fluctuations should be recognized as similarly problematic, a finding that is consistent with some previous observations in ambulatory care settings.[30-33]

Second, a history of hypoglycemia either during previous hospital days or up to 1 year prior to admission was a consistently strong predictor of hypoglycemia, a finding that has long been recognized among outpatients.[34] Whether this finding was indicative of use of overly aggressive antihyperglycemic regimens prior to admission or driven by idiosyncratic patient characteristics is unclear; regardless, this finding emphasizes the need to ascertain information on previous hypoglycemia to inform decisions regarding in-hospital care. Of note, patient age was not associated with hypoglycemia when all other risk factors were accounted for, but we found inverse linear relationships between hypoglycemia risk and body weight, albumin level, and CL_{cr}; those factors can be assessed to identify an easily recognizable "frailty phenotype" that puts patients at high risk for hypoglycemia.[35]

Third, we characterized 4 types of antihyperglycemic medications whose use independently contributed to increased hypoglycemic risk: insulin infusions, long-acting formulations, high-dose subcutaneous insulin, and sulfonylureas. Differences between insulin formulations were not considered because of restrictions in study hospital formularies and sample size constraints. After accounting for these high-risk regimens, general use of insulin (e.g., use of sliding-scale insulin) was protective against hypoglycemia. That finding was consistent with previous research indicating that orders for sliding-scale insulin, usually intended to regulate extreme hyperglycemia, rarely specify normoglycemic targets and thus pose less hypoglycemic risk than other types of insulin orders.[36]

Finally, using risk factors that we specifically created to capture scenarios of potentially inadequate glucose management, we had varying success in predicting hypoglycemia. For example, transition between hospital units requires that clinicians perform medication reconciliation, and their failure to do so may result in adverse drug events. We found such transitions to be protective against hypoglycemia, which may indicate that there were delays in resuming antihyperglycemic therapy or problems in maintaining glucose control when switching from insulin infusions in an ICU to a subcutaneous regimen in an acute care unit. Increased BG monitoring, modeled as a higher number of BG levels measured per day, was found to have a positive correlation with hypoglycemia, a finding that was likely indicative of the use of more aggressive insulin regimens. Notably, we found a positive association between hypoglycemia and failure to monitor BG before bedtime. Factors suggesting failure to consider reduced food intake, as can occur in patients who experience nausea or have active orders not to receive any food or medications orally, did not act as significant predictors, which may be indicative of adequate preventive protocols in the study hospitals. Concomitant use of hypoglycemic drugs with no indication for treatment of hyperglycemia was weakly associated with hypoglycemia events, but sample size constraints limited our ability to examine the potential contributing roles of individual drugs. With regard to interpreting these reported associations, we caution that local practice patterns may be strongly influential, as the evaluated variables reflect not only hypoglycemic risk itself but also how well clinicians are already addressing such risk.

The study had several strengths and limitations. First, our models were purposely targeted to patients who receive antihyperglycemic treatment in an effort to focus potential interventions on drug regimen changes. Beyond this restriction, all patients were included regardless of regimen type and reason for hyperglycemia. Second, our models use data that are available from discrete fields in commonly employed EHR systems. Use of EHR data allowed assessment of more than 60 hypoglycemia risk factors, but it also confined the scope of our models to variables that are collected in routine clinical care and that can be automatically employed in the risk models. While this approach allows broad implementation in theory, local capacity to extract and curate data is needed to provide a solid foundation for risk model implementation. Importantly, as new data elements are added to the EHR or value sets are modified, implementation of our algorithm must be paralleled by a maintenance plan that provides for updating information (e.g., adding new drugs) and the model itself as needed. Future efforts could also expand data ascertainment to text documents through the use of natural language processing, but data curation efforts and local variability would likely increase, necessitating model adjustments.

Third, our models were designed to allow flexibility in selection and specification of risk factor associations as patients progress during their hospital stay. Indeed, we found that some risk factors were strong predictors on admission but lost their predictive value during follow-up or vice versa. We paid careful attention to accurate characterization of associations between risk factors and hypoglycemia, using various transformations and introducing missing indicators when missingness was informative. We assumed that variables and their association with hypoglycemia would stay constant after hospital day 5, as acuity has oftentimes normalized by then and/or patients have been discharged. Larger sample sizes are needed to test these assumptions.

Finally, our models reflect the variables "as present" in the 2 hospitals. In other words, local practice created unique meaning for some of the variables—meaning that was dependent on specific protocols, formularies, and staffing. This uniqueness implies that the predictive strength of the evaluated risk factors was dependent on the level of effort already in place to prevent hypoglycemia. For example, the observation that history of hypoglycemia lost some of its predictive ability during later hospital days may reflect appropriate adjustments after clinicians realized that an initial antihyperglycemic regimen may have been too aggressive. Likewise, the volume and accuracy of data on risk factors obtained prior to admission depend on the size and interconnectivity of the health system. As a consequence, the high specificity of our models paralleled the specific processes and data availability in both institutions. Implementation of these models in other institutions will require calibration of risk factor definitions and their association with hypoglycemia to maintain the accuracy and discriminatory power of the risk score.

Conclusion

The developed prediction models were found to have excellent discriminative validity and good calibration, allowing clinicians to focus interventions on a select high-risk population in which the majority of hypoglycemic events occur.

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Disclosures

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Additional information

This article reflects the views of the authors and should not be construed to represent the views or policies of the Food and Drug Administration.

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Appendix A—Antidiabetic agents considered in study, by class

Amylinomimetic: pramlintide
Biguanide: metformin
Dipeptidyl peptidase IV inhibitors: exenatide, incretin mimetics, linagliptin, liraglutide, saxagliptin, and sitagliptin
Insulins:

- Long-acting: insulin aspart protamine with insulin aspart (70/30), insulin detemir, insulin glargine, insulin lispro protamine with insulin lispro (75/25), neutral protamine Hagedorn (NPH) insulin, and NPH with insulin regular
- Short-acting: insulin aspart, insulin glulisine, insulin human regular (not in i.v. bag), and insulin lispro
- Others: insulin human regular in i.v. infusion bag (administered via continuous infusion), and insulin zinc

Meglitinides: chlorpropamide, glimepiride, glyburide, nateglinide, pioglitazone, repaglinide, rosiglitazone, sulfonylureas, tolazamide, tolbutamide, and thiazolidinediones

Appendix B—Definitions of candidate predictors of hypoglycemia risk[a]

Candidate Predictor	Variable Type	Definition and Data Collection	Look-Back Period
Basic Characteristics			
Sex	Binary	Male sex	None
Age	Continuous	Age (yr) on admission	None
Body weight	Continuous	Most recent body weight (kg)	1 mo prior to admission
Body temperature	Continuous	Most recent body temperature (°C) on risk model day	None
Treatment in ICU	Binary	Administration of antidiabetic drug in ICU on risk model day	None
BG Values and History of Hypoglycemia			
Minimum BG value	Continuous	Lowest BG value on risk model day; if missing, impute as 120 mg/dL	None
Maximum BG value	Continuous (modeled as restricted cubic spline)	Highest BG value on risk model day; if missing, impute as 120 mg/dL	None
Admission BG	Continuous (modeled as restricted cubic spline)	First BG value on admission; if missing, impute as 120 mg/dL	On admission
BG fluctuation, log transformed	Continuous	Log of difference between BG maximum and minimum values on risk model day; if less than 2 values available, impute as mean of nonmissing values across modeling population	None
BG trend	Continuous (modeled as restricted cubic spline)	Slope of least-squares regression line for most recent BG values obtained up to 4 days prior to risk model day; to calculate slope, minimum of 4 values required (up to 10 values may be used); if fewer than 4 BG values available, use mean of nonmissing values across modeling population	4 days
No. prior days with BG of <70 mg/dL	Categorical (range, 0–5)	Count of prior days, including risk model day, with BG of <70 mg/dL	4 days
History of hypoglycemia	Binary	Presence of BG value of <70 mg/dL or hypoglycemia-related ICD-9-CM codes (251.0: hypoglycemic coma, 251.1: other specified hypoglycemia, or 251.2: hypoglycemia, unspecified) in records of inpatient or outpatient encounters within 365 days prior to current admission	365 days prior to admission
In-Hospital Use of Antihyperglycemic Drugs			
Insulin (any type)	Binary	Administration of any type of insulin on risk model day	None
Insulin infusion	Binary	Active insulin infusion order on risk model day	None
High-dose subcutaneous insulin	Binary	Administration of subcutaneous insulin on risk model day or previous day with insulin dose of >1 unit/kg/day	1 day
Long-acting insulin	Binary	Use of long-acting insulin on risk model day	None
Sulfonylureas	Binary	Use of sulfonylureas on risk model day	None

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Candidate Predictor	Variable Type	Definition and Data Collection	Look-Back Period
New initiation of antihyperglycemic agent	Binary	Use of antihyperglycemic agent indicated for treatment of hyperglycemia on risk model day that was not administered on previous risk model days	1 day
No. nonantidiabetic agents with hypoglycemic effects	Binary	Count of unique agents not indicated for treatment of hyperglycemia that can cause hypoglycemia (including aripiprazole, azithromycin, carvedilol, codeine, doxepin, gabapentin, haloperidol, hydromorphone, ibuprofen, iron, lanreotide, lorcaserin, mecasermin, metreleptin, micafungin, pancrelipase, pasireotide, pentamidine, piperacillin, pregabalin, quetiapine, sunitinib, tromethamine, and venlafaxine) used on risk model day	None
Change in antidiabetic regimen	Binary	Changes in antihyperglycemic therapy on risk model day relative to previous day, as follows: (1) addition or discontinuation of insulin infusion after subcutaneous insulin use, (2) change from sliding-scale insulin only to basal–bolus insulin therapy with or without sliding-scale insulin, and (3) change from oral antidiabetic medication only to any type of insulin with or without oral antidiabetic medication (or vice versa)	1 day
BG Monitoring			
Insufficient BG monitoring for long-acting insulin regimen	Binary	Use of long-acting insulin after 8 a.m. with no BG value recorded by 6 p.m. on risk model day	None
No. recorded BG values	Continuous	Count of BG values recorded on risk model day; counts of >10 truncated to 10	None
Change in Oral Intake			
Nausea	Binary	Administration of antiemetic drugs (including aprepitant, dolasetron, dronabinol, fosaprepitant, granisetron, metoclopramide, nabilone, ondansetron, palonosetron, prochlorperazine, promethazine, and trimethobenzamide) on risk model day	None
Nothing-by-mouth order	Binary	Presence of active order on risk model day	None
Insulin use in presence of nothing-by-mouth order	Binary	Administration of any insulin while nothing-by-mouth order active on risk model day	None
Use of long-acting insulin in presence of nothing-by-mouth order	Binary	Administration of long-acting insulin while nothing-by-mouth order active on risk model day	None
Use of dextrose 50%	Binary	Administration of dextrose 50% on risk model day	None
Enteral feeding	Binary	Presence of active enteral feeding order on risk model day	4 days
Laboratory Values			
Serum albumin	Continuous	Most recent serum albumin value (mg/dL); if no value recorded, use missing-data indicator (missing = 1, other values = 0)	4 days
Serum albumin missing-data indicator	Binary	No serum albumin value recorded	4 days

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Candidate Predictor	Variable Type	Definition and Data Collection	Look-Back Period
Estimated CL_{cr}	Continuous	Most recent CL_{cr} value, as estimated using Cockcroft–Gault equation; if several values for 1 day available, use average SCr for calculation; if CL_{cr} value missing, use missing-data indicator (missing = 1, other values = 0); if CL_{cr} value is greater than 100 mL/min, use high-value indicator (CL_{cr} value of >100 mL/min = 1; other values, including missing value = 0)	For SCr, 30 days; for weight, 30 days from SCr measurement; for height, 365 days from SCr measurement
Estimated CL_{cr} of >100 mL/min	Binary	Most recent CL_{cr} value, as estimated using Cockcroft–Gault equation, of >100 mL/min	None
CL_{cr} missing-data indicator	Binary	CL_{cr} cannot be estimated due to missing values for SCr, weight, and/or height	None
Other Variables			
Ischemic heart disease	Binary	Presence of ischemic heart disease–related ICD-9-CM codes (410.xx: acute myocardial infarction, 411.xx: other acute and subacute forms of ischemic heart disease, 412.xx: old myocardial infarction, 414.xx: other forms of chronic ischemic heart disease, V45.81: postsurgical aortocoronary bypass status, or V45.82: percutaneous transluminal coronary angioplasty status) in records of any previous inpatient or outpatient encounter	365 days prior to admission
Type 1 diabetes mellitus	Binary	Presence of type 1 diabetes–related ICD-9-CM code (250.x1 or 250.x3) in records of any previous inpatient or outpatient encounter	365 days prior to admission
Transition between ACU and ICU	Binary	Patients transitioned from ACU to ICU (or vice versa) on risk model day	None

[a]ICU = intensive care unit; BG = blood glucose; ICD-9-CM = International Classification of Diseases, Ninth Revision, Clinical Modification; CL_{cr} = creatinine clearance; SCr = serum creatinine; ACU = acute care unit.

Appendix C—Decile distribution of hypoglycemia events and data on predicted versus observed event rate

Risk Period	Decile	Sample Size	No. Events	% of Total Events	Hypoglycemia Events per 100 Admissions	
					Predicted	Observed (95% CI)
Day 1 *n* = 12,264 admissions	1	1,227	0	0	0.00121	0.00163 (0.00041–0.00649)
	2	1,228	3	1.0	0.00267	0.00081 (0.00011–0.00576)
	3	1,226	4	1.4	0.00413	0.00326 (0.00123–0.00866)
	4	1,226	3	1.0	0.00594	0.00245 (0.00079–0.00756)
	5	1,227	6	2.0	0.00834	0.00489 (0.00220–0.01084)
	6	1,226	15	5.1	0.01165	0.01223 (0.00739–0.02019)
	7	1,226	16	5.4	0.01700	0.01305 (0.00801–0.02120)
	8	1,226	45	15.3	0.02597[a]	0.03670 (0.02751–0.04881)
	9	1,226	59	20.1	0.04390	0.04812 (0.03746–0.06162)
	10	1,226	143	48.6	0.11898	0.11664 (0.09984–0.13584)

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Risk Period	Decile	Sample Size	No. Events	% of Total Events	Hypoglycemia Events per 100 Admissions	
					Predicted	Observed (95% CI)
Day 2 n = 15,810 admissions	1	1,581	1	0.6	0.00097	0.00000 (0.00000–0.00233)
	2	1,582	1	0.3	0.00191	0.00190 (0.00061–0.00586)
	3	1,582	5	0.6	0.00294	0.00126 (0.00032–0.00504)
	4	1,582	6	1.1	0.00427	0.00253 (0.00095–0.00672)
	5	1,582	9	2.5	0.00606	0.00569 (0.00296–0.01090)
	6	1,582	17	3.1	0.00860	0.00695 (0.00385–0.01251)
	7	1,581	23	5.0	0.01270	0.01139 (0.00718–0.01800)
	8	1,581	31	8.7	0.02009	0.01961 (0.01382–0.02775)
	9	1,581	69	19.3	0.03678	0.04428 (0.03517–0.05560)
	10	1,576	211	58.9	0.13253	0.13325 (0.11735–0.15094)
Days 3–5 n = 32,699 admissions	1	3,268	0	0.2	0.00081	0.00061 (0.00015–0.00244)
	2	3,270	3	0.5	0.00149	0.00061 (0.00015–0.00244)
	3	3,268	4	0.8	0.00217	0.00153 (0.00064–0.00367)
	4	3,269	3	1.0	0.00305	0.00184 (0.00082–0.00408)
	5	3,269	9	1.5	0.00428	0.00275 (0.00143–0.00528)
	6	3,271	15	2.8	0.00622	0.00520 (0.00323–0.00834)
	7	3,271	16	3.8	0.00934	0.00703 (0.00468–0.01056)
	8	3,269	45	7.5	0.01519	0.01377 (0.01029–0.01839)
	9	3,269	114	18.9	0.02834[a]	0.03487 (0.02910–0.04174)
	10	3,264	381	63.1	0.11404	0.11673 (0.10616–0.12820)

[a]Predicted value is outside the 95% confidence interval of the observed event rate.

Appendix D—Number and locations of knots for nonlinear predictors modeled as cubic spline[a,b]

Period	Predictor	No. Knots[a]	Location of Knots				
			1st	2nd	3rd	4th	5th
Day 1	BG trend	5	–45	–2	0	3	23
Day 2	BG maximum (mg)	4	100	192	262	510	
	BG trend	5	–14	–1	0	2	12
	Admission BG (mg)	4	51	129	197	586	
Days 3–5	BG trend	5	–10	–1	0	1	11
	Admission BG (mg)	4	48	126	190	607	

[a]BG = blood glucose.

[b]Restricted cubic splines allow modeling of nonlinear associations between continuous variables and an outcome. The knots define the turning points of the fitted association. For example, BG trend in the day 1 model had different associations with the outcome across 6 segments (<–45, –45 to approximately –2, –2 to approximately 0, 0 to approximately 3, 3 to approximately 23, and >23) defined by 5 knots.